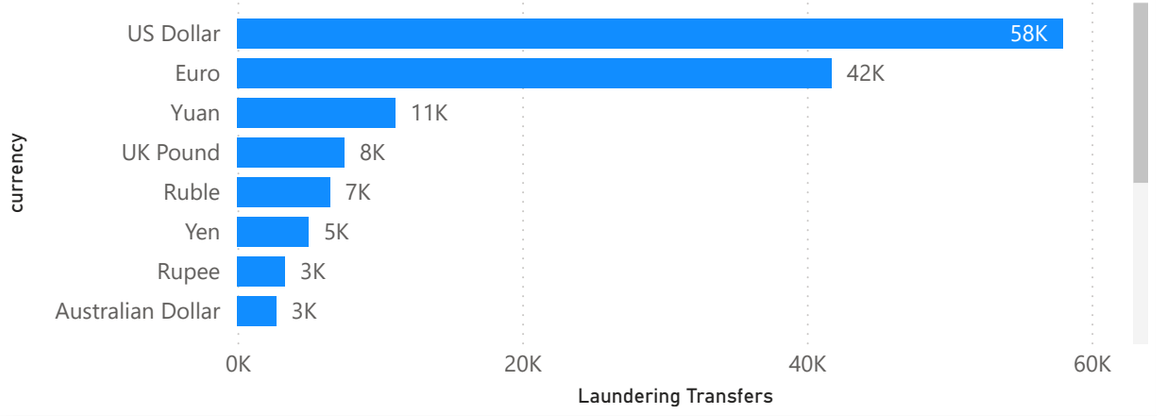
Laundering Overview: Volume by Format, Currency and Time

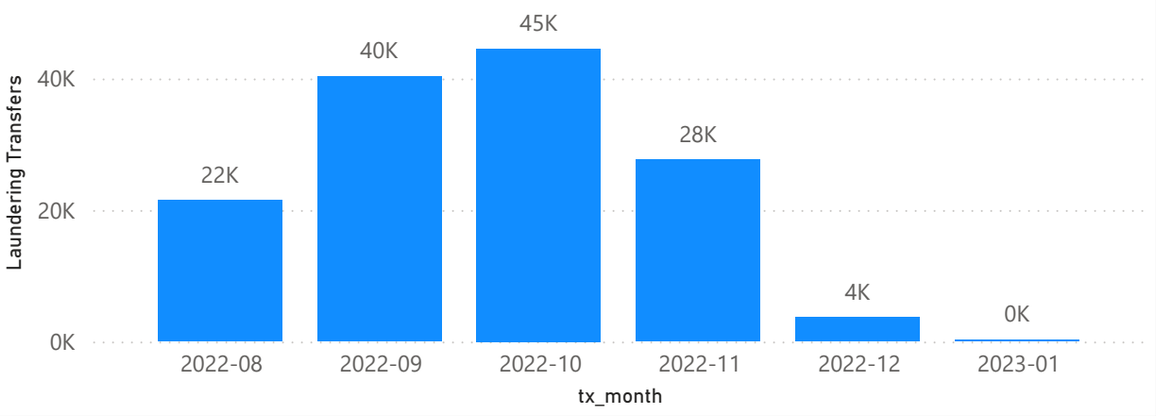
SYNTHETIC DATA: IBM AMLworld simulation (HI-Large laundering patterns) · every transfer is labelled laundering · August 2022 to January 2023

138K	16K	8.4	99.99%
Laundering Transfers	Total Schemes	Avg Transfers per Scheme	ACH Share

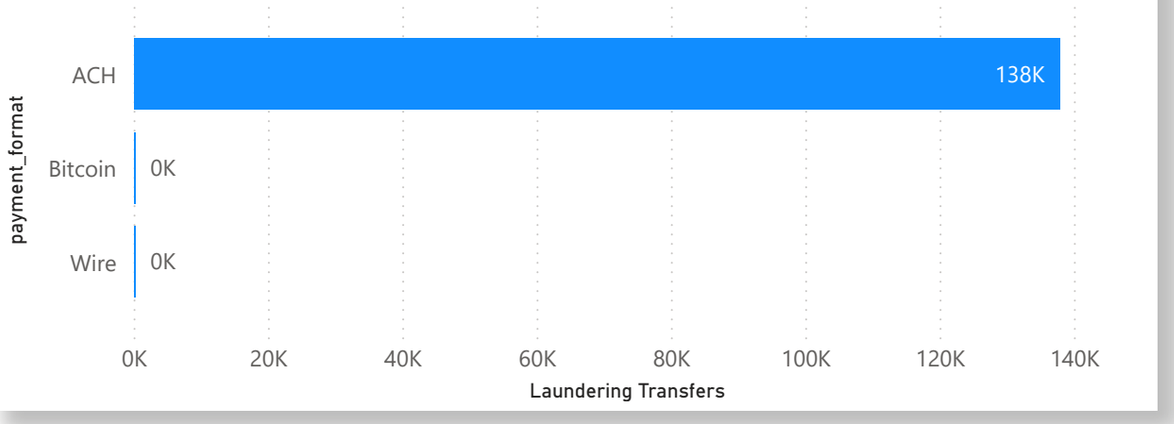
72% of laundering is in US dollars or euros



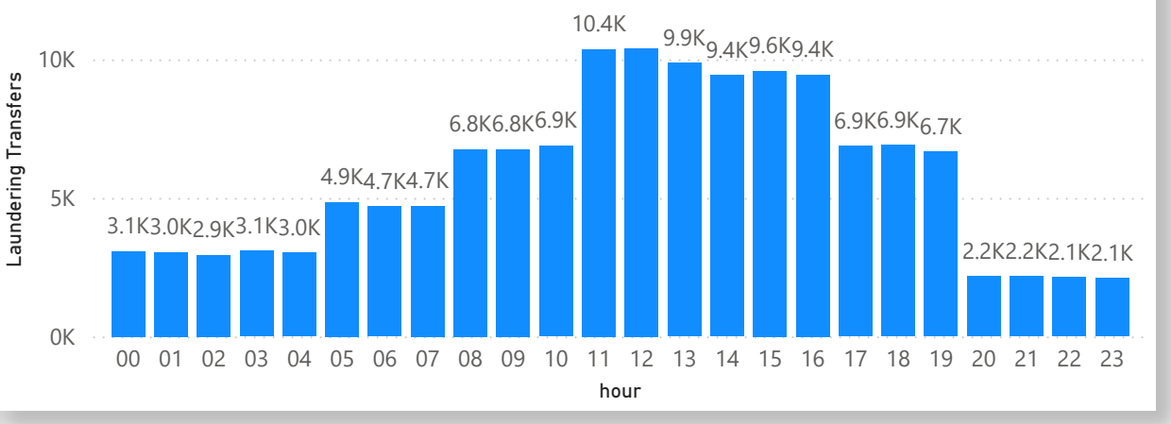
Laundering transfers by month: October 2022 was the peak



99.99% of laundering is paid by ACH



Laundering is less likely at night: 14.5% of transfers in 25% of the day



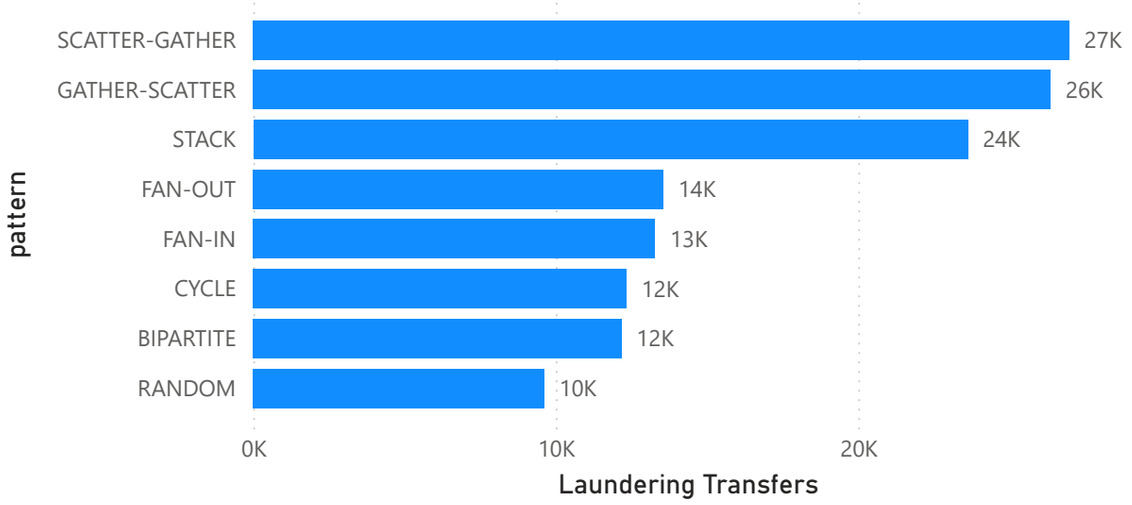
Laundering Patterns: Amount Band and Scheme Type

SYNTHETIC DATA: IBM AMLworld simulation (HI-Large laundering patterns) · a scheme is one laundering attempt, such as a fan-in or a cycle,

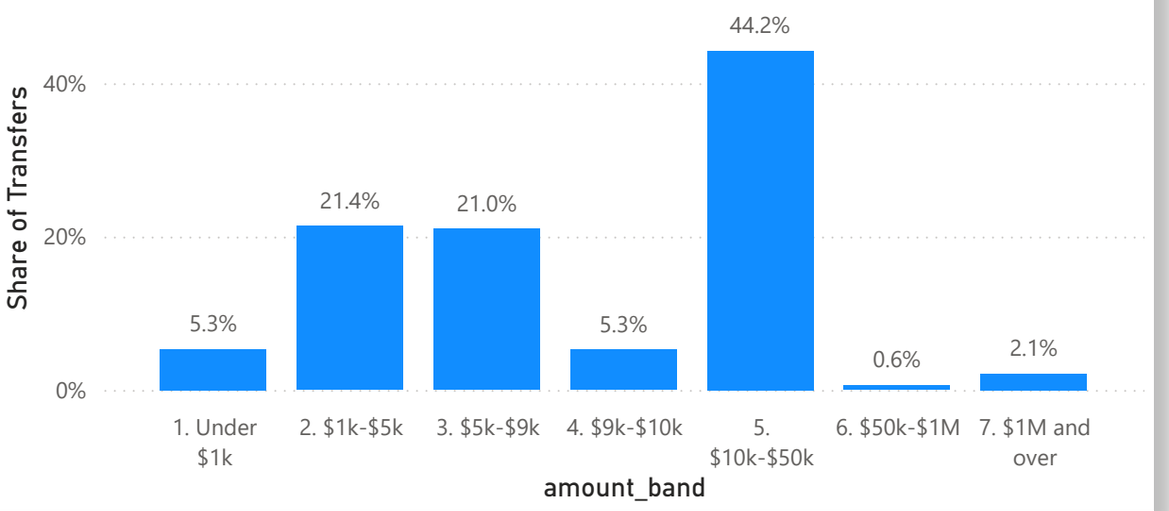
Pattern

All

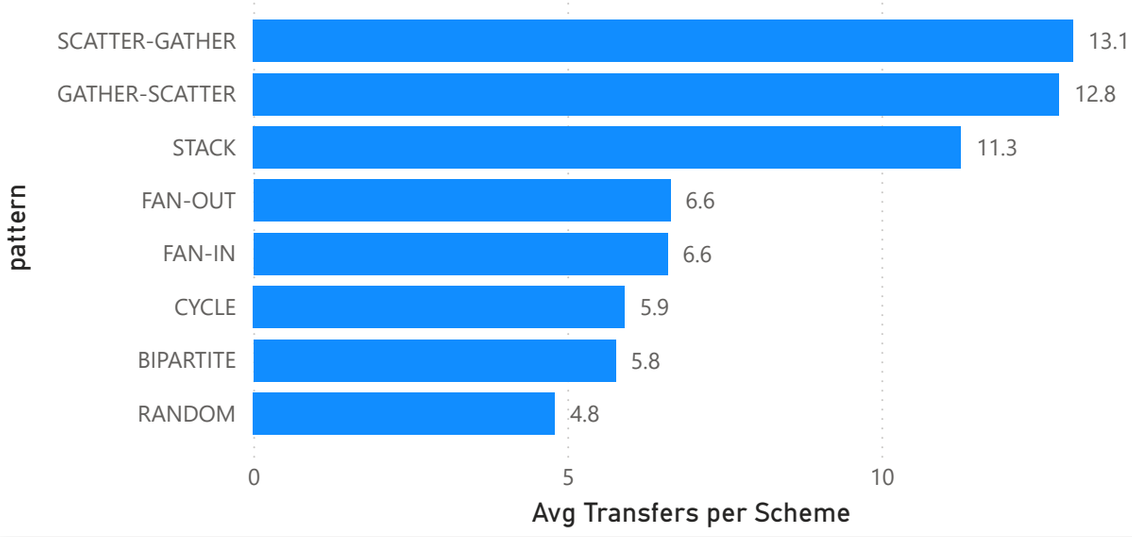
Scatter-gather and gather-scatter schemes move the most transfers



44% of laundering is $10k-$50k, with no spike just below $10k



Scatter-gather schemes are the largest, at about 13 transfers each



Every pattern is about 2,000 schemes; bipartite and random are hardest to detect

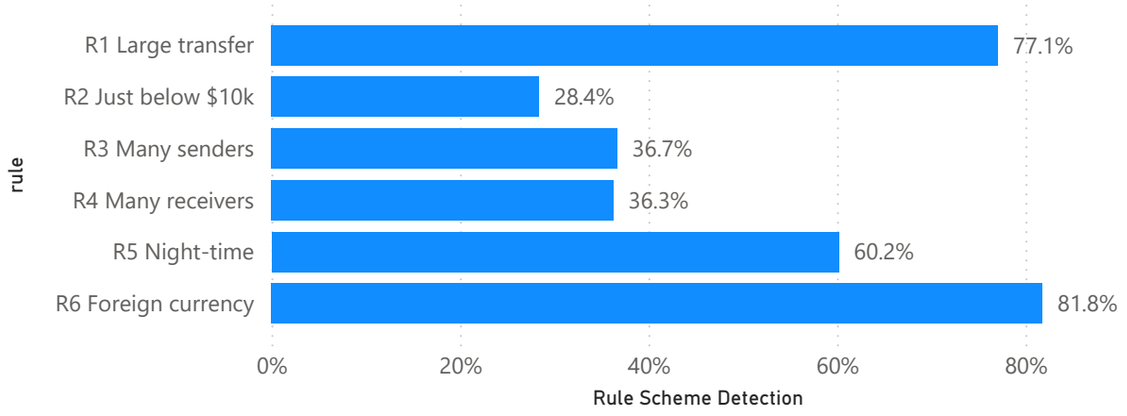
pattern	Total Schemes	Avg Transfers per Scheme	Scheme Detection Rate
BIPARTITE	2,109	5.8	95.9%
RANDOM	2,001	4.8	96.0%
FAN-IN	2,014	6.6	96.7%
FAN-OUT	2,040	6.6	97.1%
STACK	2,096	11.3	97.4%
CYCLE	2,086	5.9	97.5%
SCATTER-GATHER	2,067	13.1	97.8%
GATHER-SCATTER	2,054	12.8	99.3%
Total	**16,467**	**8.4**	**97.2%**

Rule Performance: What Each Monitoring Rule Catches

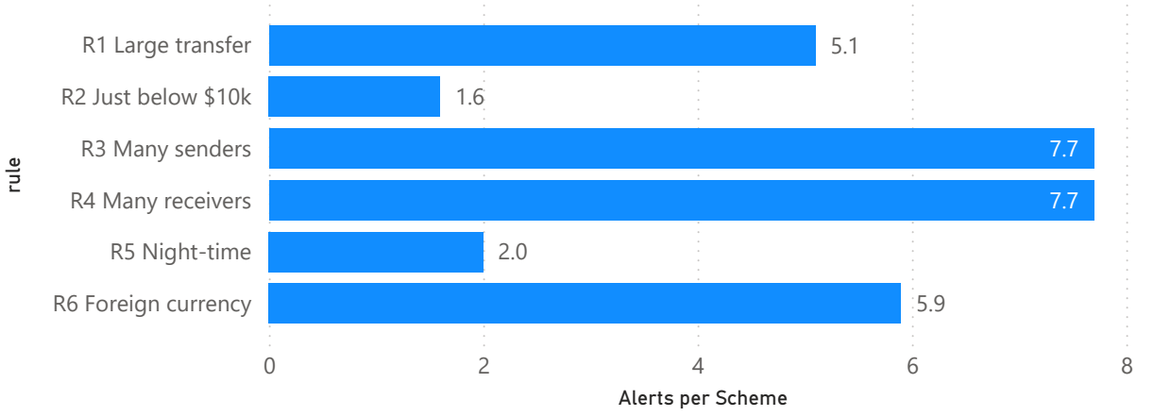
SYNTHETIC DATA: IBM AMLworld simulation (HI-Large laundering patterns) · six simple rules run on every laundering transfer · a scheme is

265K	97.2%	459	16.5
Rule Alerts	Scheme Detection Rate	Schemes Missed	Alerts per Scheme Detected

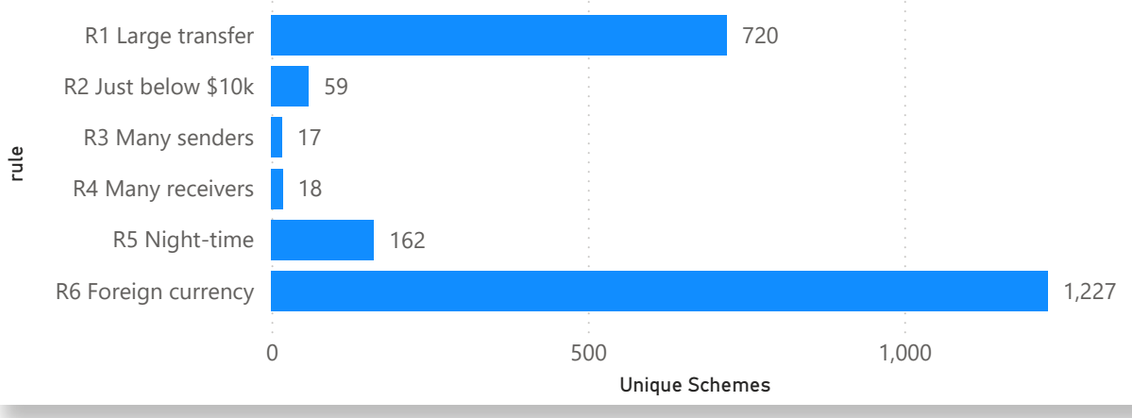
R6 (foreign currency) and R1 (large transfer) detect the most schemes



R3 and R4 need 8.3 alerts per scheme: the heaviest workload



R3 and R4 catch only 17 schemes each that no other rule catches



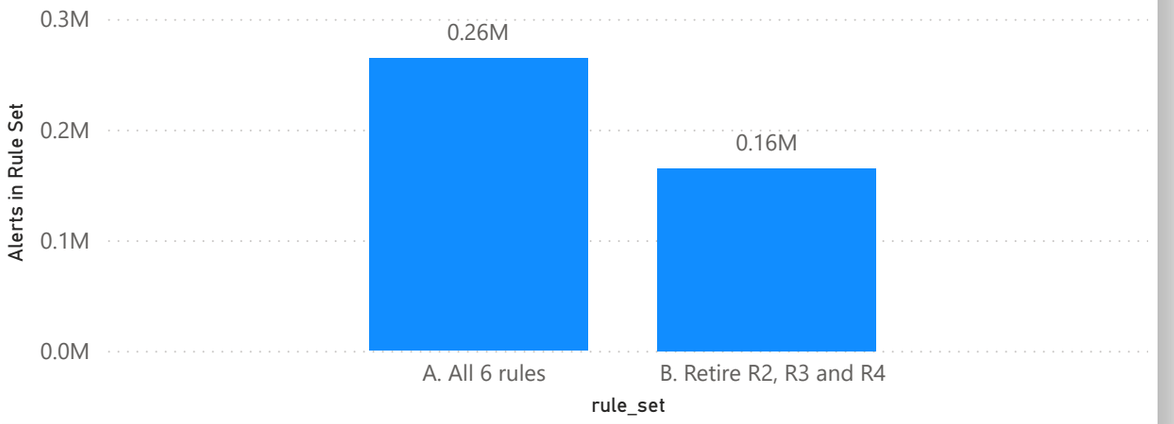
% of schemes detected by each rule, by pattern

pattern	r1_large	r2_below_10k	r3_many_senders	r4_many_receivers	r5_night	r6_foreig
BIPARTITE	80.0	25.0	23.0	22.0	45.0	83
RANDOM	52.0	13.0	18.0	17.0	52.0	83
FAN-IN	84.0	29.0	61.0	21.0	59.0	59
FAN-OUT	84.0	26.0	23.0	61.0	58.0	86
STACK	81.0	33.0	26.0	26.0	62.0	86
CYCLE	56.0	17.0	19.0	20.0	57.0	85
SCATTER-GATHER	86.0	41.0	61.0	62.0	72.0	85
GATHER-SCATTER	93.0	43.0	63.0	62.0	77.0	86

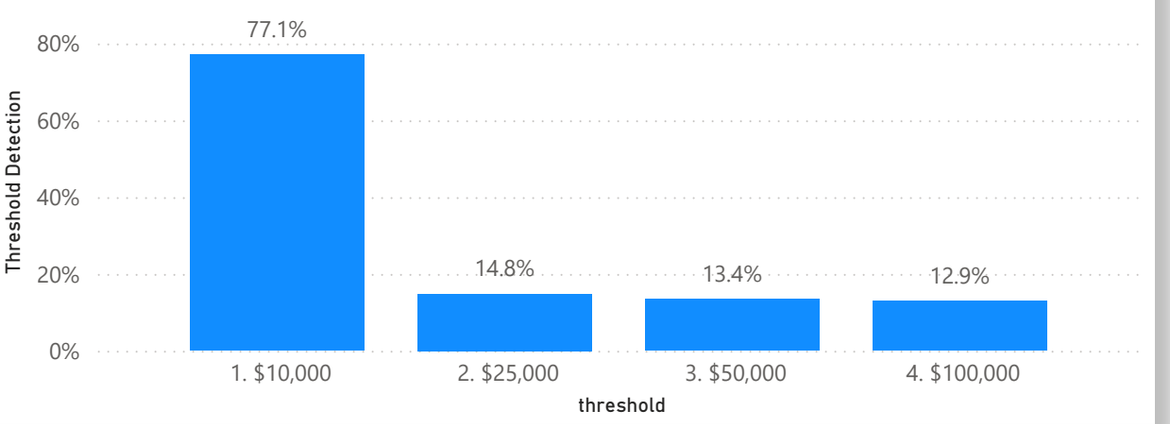
Rule Tuning Plan: Fewer Alerts, Same Coverage

For a financial crime operations lead · built on synthetic data from the IBM AMLworld simulation

Retiring R2, R3 and R4 cuts alerts by 37% and detection by under 1 point



Do not tighten R1: at $25k it would detect 15% of schemes, not 77%



Recommended tuning plan

1. Retire R3 and R4 (many senders, many receivers): 8.3 alerts per scheme, yet only 17 schemes each that no other rule catches.
2. Retire R2 (just below $10k): there is no structuring spike below $10,000, and it catches only 59 unique schemes.
3. Keep R1 at $10,000: a $25,000 threshold would cut its scheme detection from 77% to 15%.
4. Test R5 (night-time) and R6 (foreign currency) on the full transactions file: they are broad rules that may also flag many normal payments.

Result: 37% fewer alerts (262,703 to 164,775) for 0.9 points less scheme detection (97.2% to 96.3%).

KPIs to track

Scheme detection rate
Target: 96% or higher
Alerts per scheme detected
Target: down by a third after tuning
False-alert rate on normal payments
Measure on the full file before keeping R5 and R6

Caveats

- Synthetic data: no real people, banks or accounts.
- This file holds only laundering, so false alerts cannot be measured here.